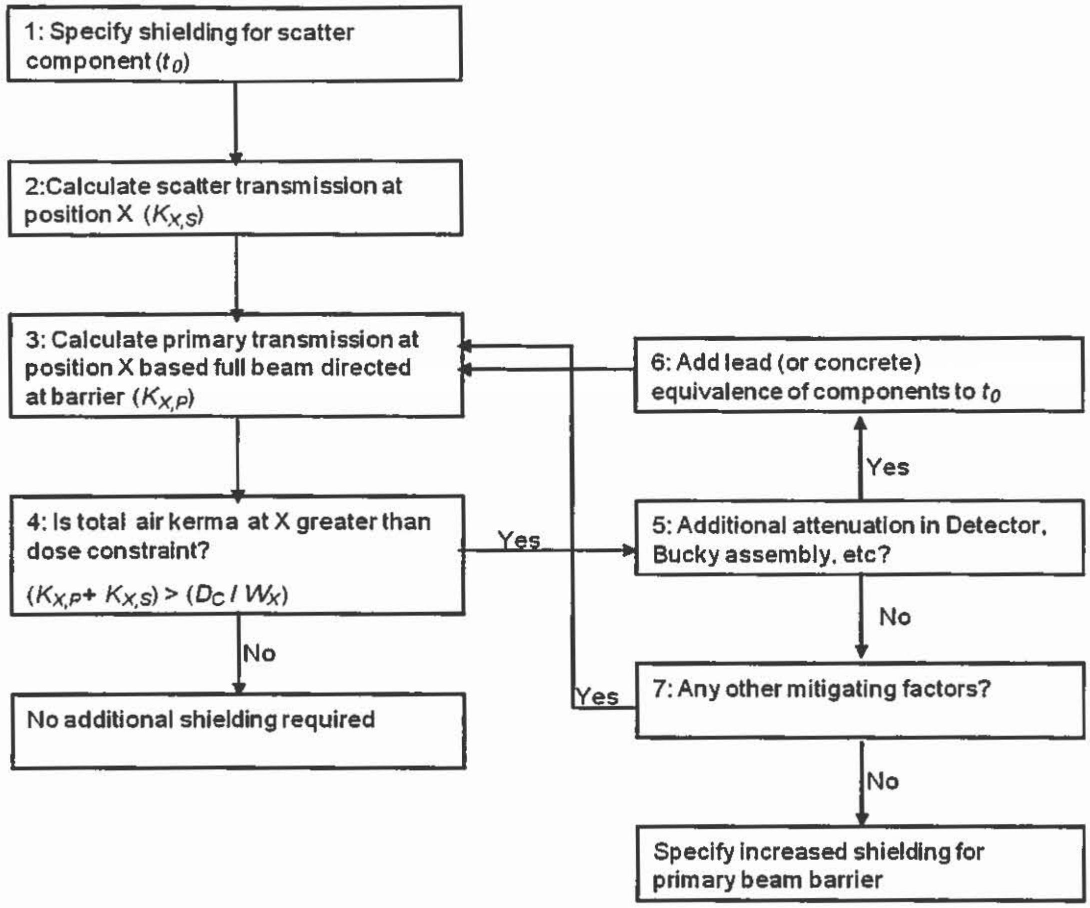

Figure 2.5 Recommended method for specification of primary shielding for wall bucky in radiography.

2.4 Tertiary scatter from ceilings and walls

Although air kerma levels from tertiary scatter off solid ceiling slabs and walls are only a few percent of those from direct exposure, they can become significant when workloads are high. Protection against tertiary scatter will almost always need to be provided for current multislice helical CT scanners and may need to be considered for interventional suites with heavy workloads (see Figure 2.6). Dose levels are strongly dependent on the height of the ceiling and that of the protective barrier. This is illustrated by a plot of scatter level *versus* ceiling height for different barrier heights in interventional and CT scanner rooms (see Figure 2.7). A methodology employing a linear equation has been developed to allow scatter levels to be estimated for standard configurations representing a CT room with a scanner operating at 120 kV and an interventional radiology or cardiology room with an average tube potential of 85 kV (Martin et al, 2012). The amount of radiation scattered from walls increases substantially with tube potential, so choice of a higher tube potential represents a conservative approach.

The method employs an equation to derive tertiary scatter S_{ceiling} of the form:

$$S_{\text{ceiling}} = (C - m \times b) \times D \tag{2.4}$$

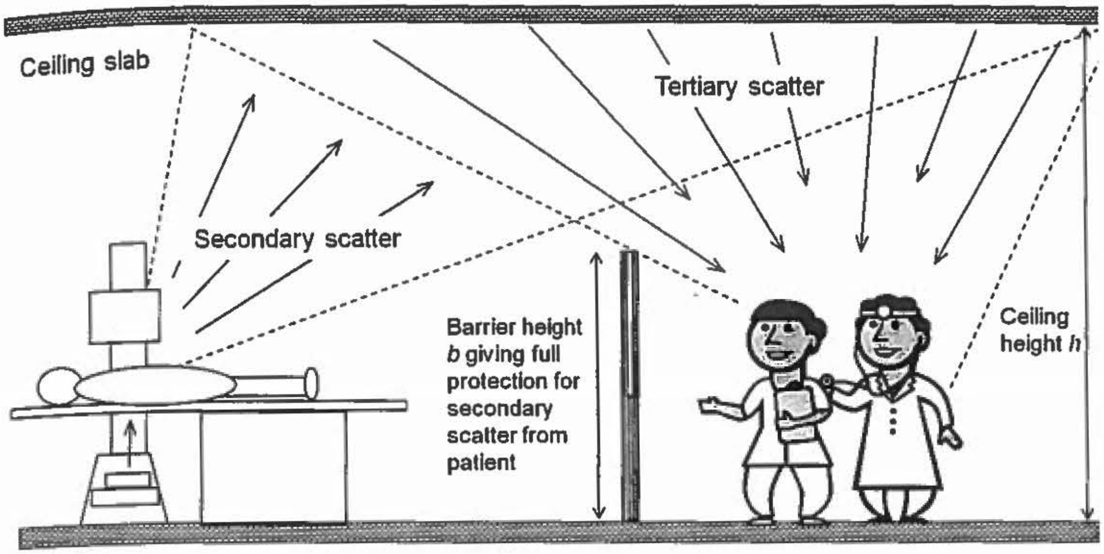

Figure 2.6 How tertiary scatter from ceilings can contribute to staff dose.

where b is the height of the barrier between the staff and X-ray equipment, D is the relevant dose quantity and m and C are constants for a particular ceiling height. The majority of the tertiary scatter contributing to personnel dose results from a 3 m wide strip of ceiling slab between the X-ray unit and the operator or other personnel. Therefore, it is possible to use standard factors in the equation linked simply to the ceiling height. Values of constants relating to a CT scanner room and an interventional suite are given in Table 2.5. The factors have been derived so that they can be applied directly to the patient dose quantities used for estimating scatter. For CT scanning, this would be the DLP workload with half the DLP for the head being added to that of the body, while for an interventional suite, this would be the KAP. Examples of the application of the method are given in the relevant chapters.

Tertiary scatter from a wall adjacent to an unshielded entrance to a control cubicle (S_{wall}) can also be significant. Doors will almost always be required for CT facilities to protect against this, but protection may also need to be installed in the door of an interventional suite. The magnitude of the scatter is related to the perpendicular distance, d (measured in metres), between the patient/isocentre and the wall adjacent to the entrance. Levels of tertiary scatter at the edge of a protected screen can be predicted from the KAP in gray centimetres squared for an interventional suite using the equation:

$$S_{wall} = \mathrm{KAP} \times 0.058\mathrm{e}^{-0.56d} \tag{2.5}$$

and from the DLP in gray centimetres for a CT suite from the equation:

$$S_{wall} = \mathrm{DLP} \times 6.6\mathrm{e}^{-0.56d} \tag{2.6}$$

a)

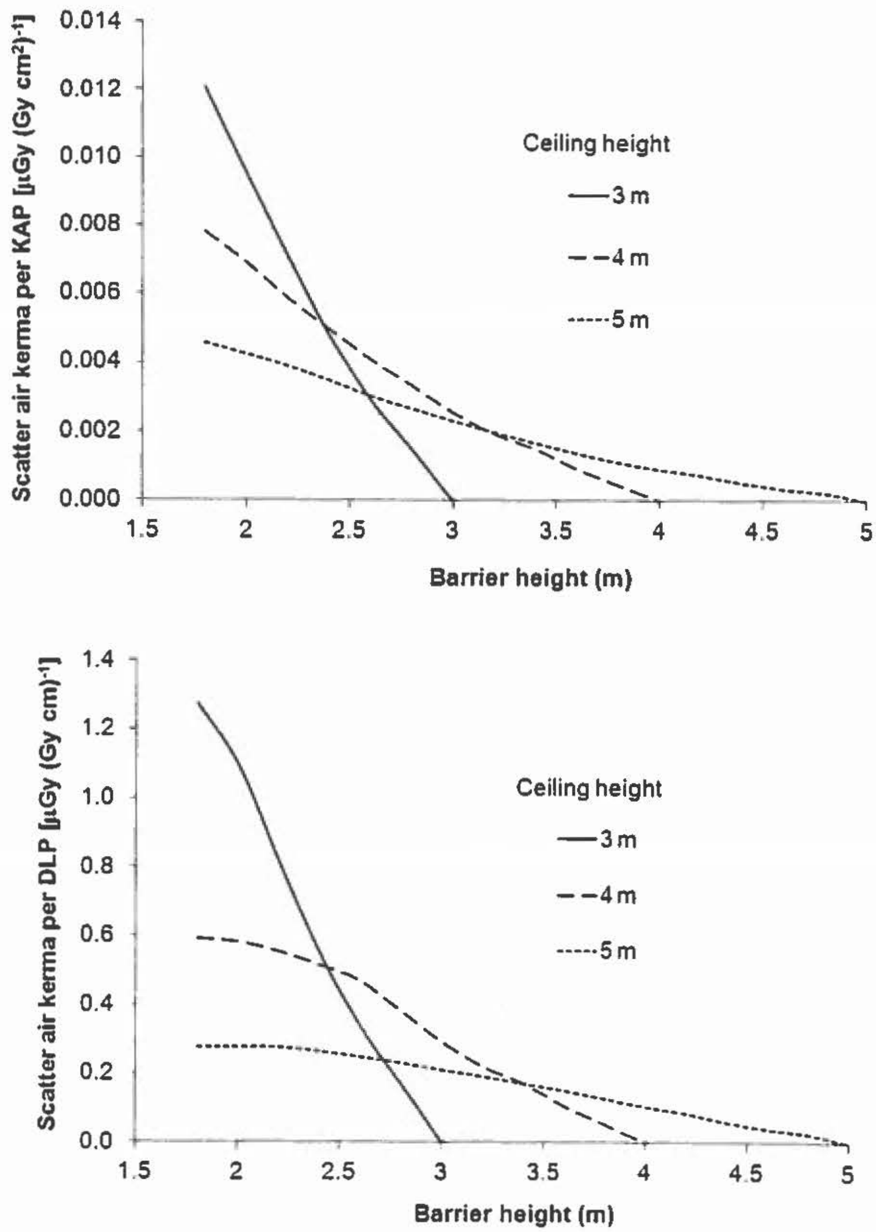

Figure 2.7 Variation of tertiary scatter air kerma incident on an operator 1.5 m from a barrier as a function of ceiling height in (a) an interventional suite and (b) a CT room. Scatter kerma is per-patient KAP in (a) and per-patient DLP in (b).

Table 2.5 Values for the coefficients based on room ceiling height that can be substituted into Equation 2.4 to predict levels of tertiary scatter air kerma from the ceiling incident on persons behind protective barriers of different heights

Ceiling height (m)	3.0	3.2	3.4	3.6	3.8	4.0	4.2	4.4	4.6	4.8	5.0
Coefficient[a]	Value										
Fluoroscopy/radiography 85 kV											
m_{KAP} [μGy (Gy cm^2)$^{-1}$]m^{-1}	0.0101	0.0081	0.0066	0.0053	0.0044	0.0036	0.0030	0.0024	0.0020	0.0017	0.0015
C_{KAP} [μGy (Gy cm^2)$^{-1}$]	0.0297	0.0254	0.0217	0.0186	0.0160	0.0137	0.0119	0.0103	0.0090	0.0078	0.0069
CT 120 kV											
m_{DLP} [μGy (Gy cm)$^{-1}$]m^{-1}	1.10	0.836	0.636	0.490	0.382	0.296	0.234	0.180	0.143	0.114	0.093
C_{DLP} [μGy (Gy cm)$^{-1}$]	3.24	2.64	2.14	1.75	1.44	1.19	0.988	0.801	0.669	0.562	0.478

[a] *m* and *C* are constants for a particular ceiling height.